



TECNOLÓGICO NACIONAL DE MÉXICO

INSTITUTO TECNOLÓGICO DE SALTILLO

Materia: Cómputo en la nube

Hora clase: 7:00am - 8:00am

4.2 Demostrar la inyección SQL

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1. Crear la base de datos

```
MariaDB [(none)]> show databases;
+-----+
| Database |
+-----+
| cnube    |
| information_schema |
| mysql    |
| performance_schema |
| sys      |
| usuarios |
| vulnerable_db |
+-----+
7 rows in set (0.004 sec)

MariaDB [(none)]> use vulnerable_db
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
MariaDB [vulnerable_db]> select * from usuarios;
+----+-----+-----+
| id | nombre_usuario | contrasena |
+----+-----+-----+
| 1  | admin          | 1234       |
| 2  | user           | password   |
+----+-----+-----+
2 rows in set (0.001 sec)

MariaDB [vulnerable_db]>
```

2. Instalación apache y php

```
cesar@svr-sql:/var/www/html$ sudo apt install php libapache2-mod-php php-mysql -y
sudo: unable to resolve host svr-sql: Name or service not known
php is already the newest version (2:8.4+96).
libapache2-mod-php is already the newest version (2:8.4+96).
php-mysql is already the newest version (2:8.4+96).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 56
```

3. Creación archivo login.php

```
cesar@svr-sql:/var/www/html$ cat login.php
<?php
$conn = new mysqli('localhost', 'root', '', 'vulnerable_db');

if ($conn->connect_error) {
    die("Connection failed: " . $conn->connect_error);
}

if ($_SERVER['REQUEST_METHOD'] == 'POST') {
    $usuario = $_POST['usuario'];
    $contrasena = $_POST['contrasena'];

    // Consulta vulnerable (SQL Injection)
    $sql = "SELECT * FROM usuarios WHERE nombre_usuario='$usuario' AND contrasena='$contrasena'";

    $result = $conn->query($sql);

    if ($result->num_rows > 0) {
        echo "Login exitoso";
    } else {
        echo "Credenciales inválidas";
    }
}
?>

<form method="POST" action="login.php">
    Usuario: <input type="text" name="usuario"><br>
    Contraseña: <input type="password" name="contrasena"><br>
    <input type="submit" value="Ingresar">
</form>
cesar@svr-sql:/var/www/html$
```

4. Autenticación normal

🔄 No seguro 192.168.234.16/login.php

Login exitoso

Usuario:

Contraseña:

5. Inyeccion SQL

🔄 No seguro 192.168.234.16/login.php

Login exitoso

Usuario:

Contraseña: